# THE TWO-WAY LINE 京张对开

Running the Centennial Jing-Zhang Corridor in Both Directions — an AI-belt urban design

Identity · Two arrows, one track

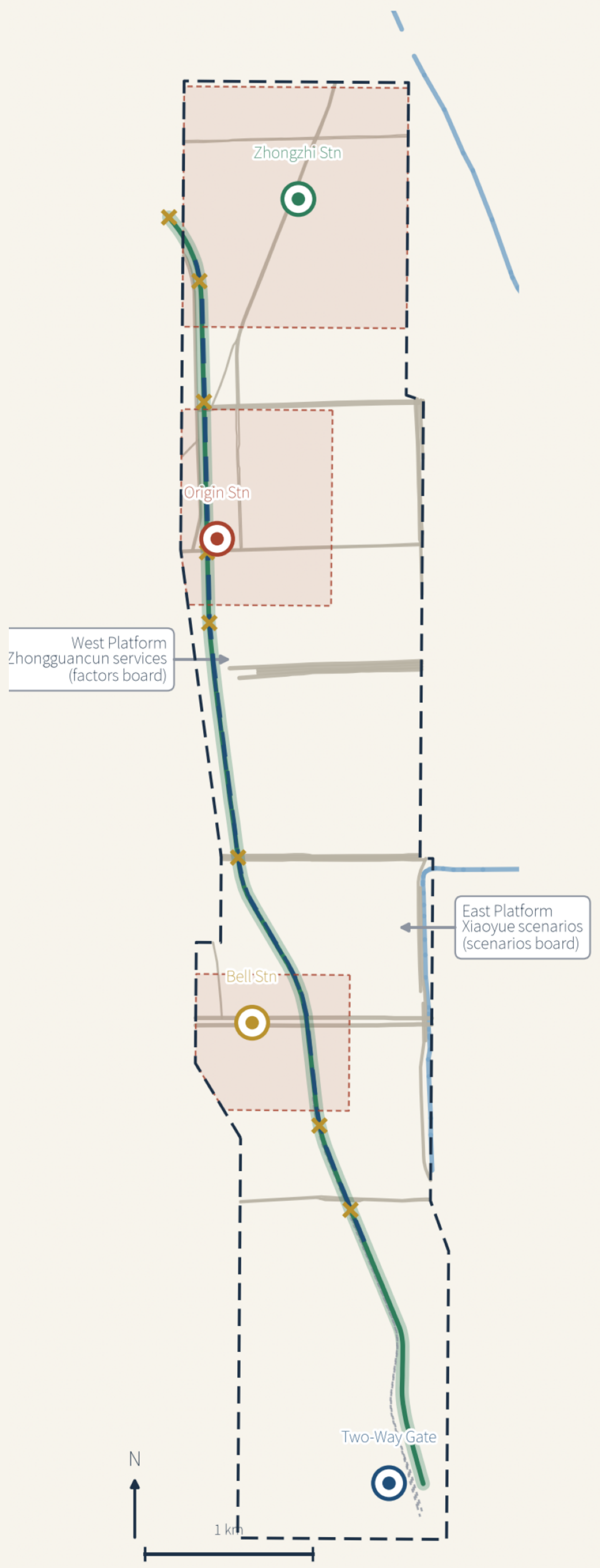
Two parallel rails bound by sleeper bars, north-pointing above and south-pointing below; the negative space between them forms the skeleton of the character 开（“open”）. Green = up-line energy and compute, deep blue = the Under Line, brick red = heritage and sleepers.

Master Concept Board

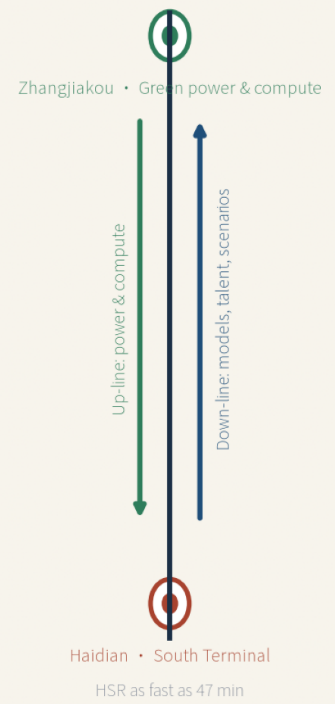
## THE TWO-WAY LINE · Master Concept

FIG-01

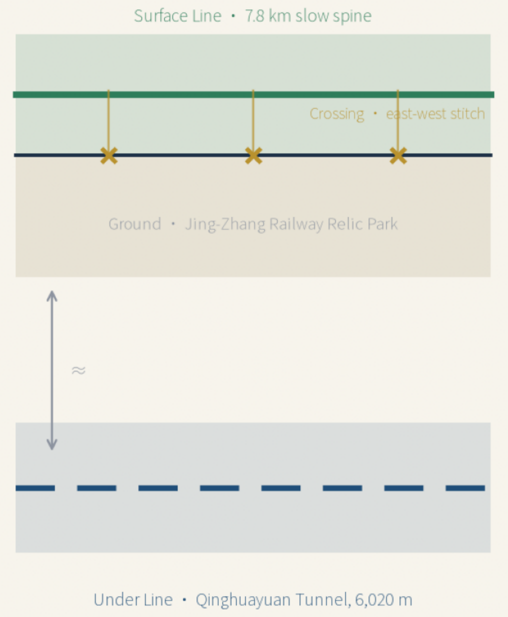
Surface + Under Lines · 3 Stations 1 Gate · 2 Platforms 8 Crossings · Twin-River Cooling Loop — 11.4 km²




The Two-Way Corridor (concept)




Surface / Under Line section (conceptual, not an engineering section)




Tunnel depth and section are indicative from public sources; no alignment or structural conclusion is expressed.

- Surface Line · 7.8 km heritage slow spine
- Under Line · Qinghuayuan HSR tunnel (indicative)
- Design area (provisional)
- Key areas (provisional)
- Crossings (8 east-west stitches)
- Qinghe / Xiaoyue R.
- Metro Line 13

### Three stations, one gate

| | |
|---|---|
| Zhongzhi<br>full-stack training & standards | 192.1 ha |
| Origin<br>campus-city switch-back | 104.3 ha |
| Bell<br>native consumption, down-line terminus | 72.0 ha |
| Gate<br>Beijing North gateway, arrival sequence | gateway |

### Eight crossings and the roads they cross

| | | |
|---|---|---|
| K1 | Xueyuan South Rd | secondary |
| K2 | North 3rd Ring | expressway |
| K3 | Zhichun Rd | arterial |
| K4 | North 4th Ring | expressway |
| K5 | Chengfu Rd | secondary |
| K6 | Qinghua East Rd | arterial |
| K7 | Yuequan Rd | branch |
| K8 | Qinghe Bay | waterfront - 85 m beyond |

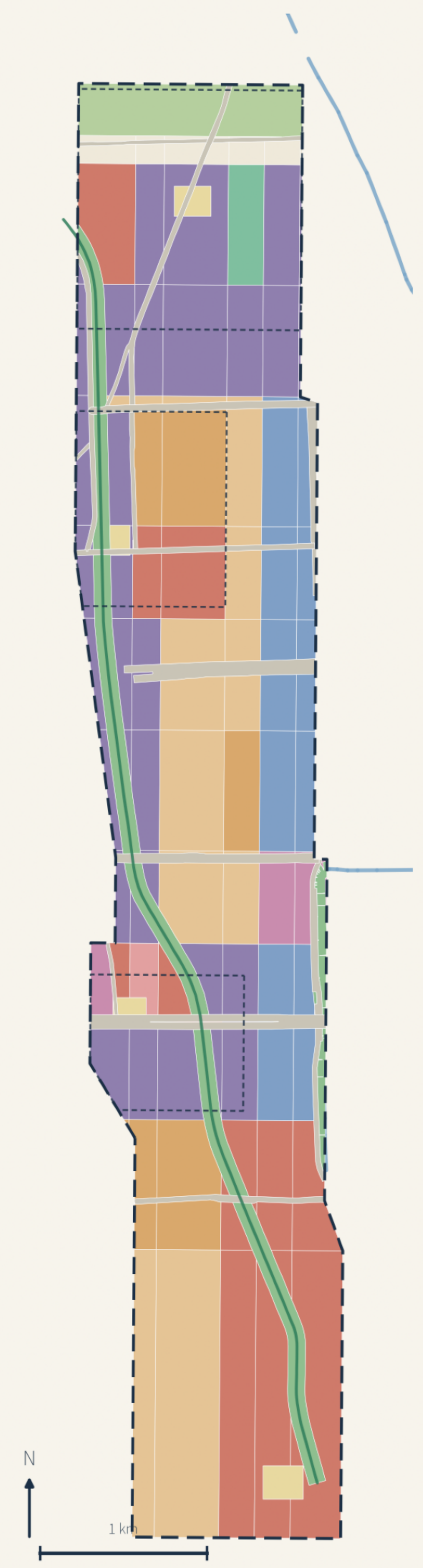
Breakpoints are located by the orthogonal road structure: the two expressways are the hardest breaks; Zhichun, Chengfu and Qinghua East carry the densest campus-city flows. K8 sits about 85 m beyond the northern edge of the design area - its job is precisely to carry the Surface Line past the boundary to the Qinghe waterfront.

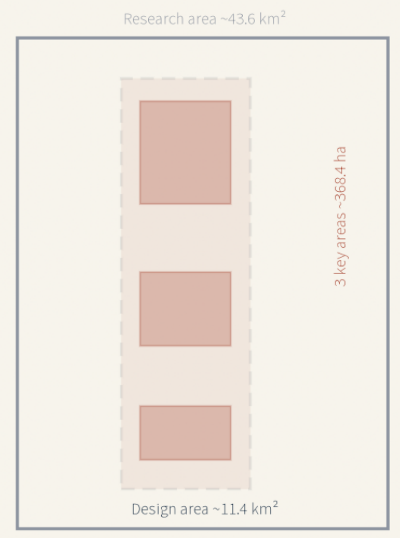
Boundary status: provisional rough frame, self-derived from OSM centerlines of announced streets and area-calibrated; not an official redline; no approval/ownership/precise-area use; full recalculation when official polygons arrive. Base data © OpenStreetMap contributors (ODbL). Derived from packaged GeoJSON + metrics; concept proposal. xr843 / jingzhang-two-way-line · 2026-08

## Three-Level Scope & Land Use

FIG-02

43.6 km² research · 11.4 km² design · 368.4 ha key areas — 189-parcel full partition




| | |
|---|---|
| Research | $258.0 \times 10^4$ m² 22.6% |
| Housing | $209.3 \times 10^4$ m² 18.4% |
| Commercial | $203.5 \times 10^4$ m² 17.9% |
| Education | $111.5 \times 10^4$ m² 9.8% |
| Community svc. | $89.9 \times 10^4$ m² 7.9% |
| Park green | $80.3 \times 10^4$ m² 7.0% |
| Roads | $74.1 \times 10^4$ m² 6.5% |
| Protective green | $40.6 \times 10^4$ m² 3.6% |
| Culture | $20.8 \times 10^4$ m² 1.8% |
| White land (reserve) | $18.6 \times 10^4$ m² 1.6% |
| Sports | $15.4 \times 10^4$ m² 1.4% |
| Plaza | $11.7 \times 10^4$ m² 1.0% |
| Health | $6.3 \times 10^4$ m² 0.6% |

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### Core thesis

Read 'Jing-Zhang' in full — the corridor has two ends. In 1909 the railway carried energy and engineering confidence into Beijing; in 2019 the HSR kept racing beneath the heritage park through Qinghuayuan Tunnel; in 2022 'Eastern Data, Western Computing' placed training compute on Zhangjiakou's green power. The belt is not a 9-km monologue but the southern terminal of a 200-km two-way relationship: power and compute up-line, models, talent, and scenarios down-line.

### Structure · Surface+Under Lines / 3 Stations 1 Gate / 2 Platforms 8 Crossings / Cooling Loop

Surface Line = the 7.8-km heritage slow spine for people and culture; Under Line = the HSR in the tunnel for speed and reach. Three stations = Zhongzhi (training & standards), Origin (campus-city switch-back), Bell (content & data assets); one gate = the Two-Way Gate at Beijing North. West Platform = Zhongguancun services; East Platform = Xiaoyue scenarios; eight crossings stitch east and west. The Twin-River Cooling Loop joins the Qinghe and Xiaoyue with cabin waste-heat reuse.

### Institutional core · the Two-Way Protocol

Every AI deployment is a paired run: an up-line run that collects data and consumes compute must be matched by a down-line run delivering services and public returns; who can stop it and how to exit is written on every card. Extraction-only deployments are never scheduled. Releases follow an annual public Timetable.

### Three key areas · the density gradient from north to south

| Key area | Area | Buildings | Concept GFA | Concept gross PR | Leading land use | Character |
|---|---|---|---|---|---|---|
| Zhongzhi · Zhongzhiyuan | 192.1 ha | 47 | 1.448 M m² | 0.75 | research 43.1% | campus |
| Origin · AI Origin Community | 104.3 ha | 49 | 1.005 M m² | 0.96 | community 35.7% | district |
| Bell · Dazhongsi | 72.0 ha | 38 | 1.208 M m² | 1.68 | research 59.3% | urban |

Concept gross plot ratio is concept floor area divided by key-area area — not a statutory plot ratio. Land use is recomputed parcel-by-parcel against each key area in EPSG:4548 and covers each area exactly.

| | |
|---|---|
| 11.40 km² | design area (provisional, +0.0006%) |
| 368.4 ha | three key areas (±0.01%) |
| 7.8 km | through slow spine |
| 8+3+1 | 12 scenario nodes |
| 12/3 | cards / industry tests |
| 14 | renewal projects, 3 phases |

### Two scope facts · disclosed by us

(1) Of the eight crossings, Qinghe Bay sits about 85 m beyond the northern edge of the overall design area — its job is to carry the Surface Line past the boundary to the Qinghe waterfront, so crossing is functionally necessary and its land is excluded from in-area recomputation. (2) Qinghuayuan Station is about 389 m west of the area, inside the coordinated research scope, so run S8 is a deliberately cross-scope ceremonial venue. Both were found by our own recomputation and registered in assumptions.json.

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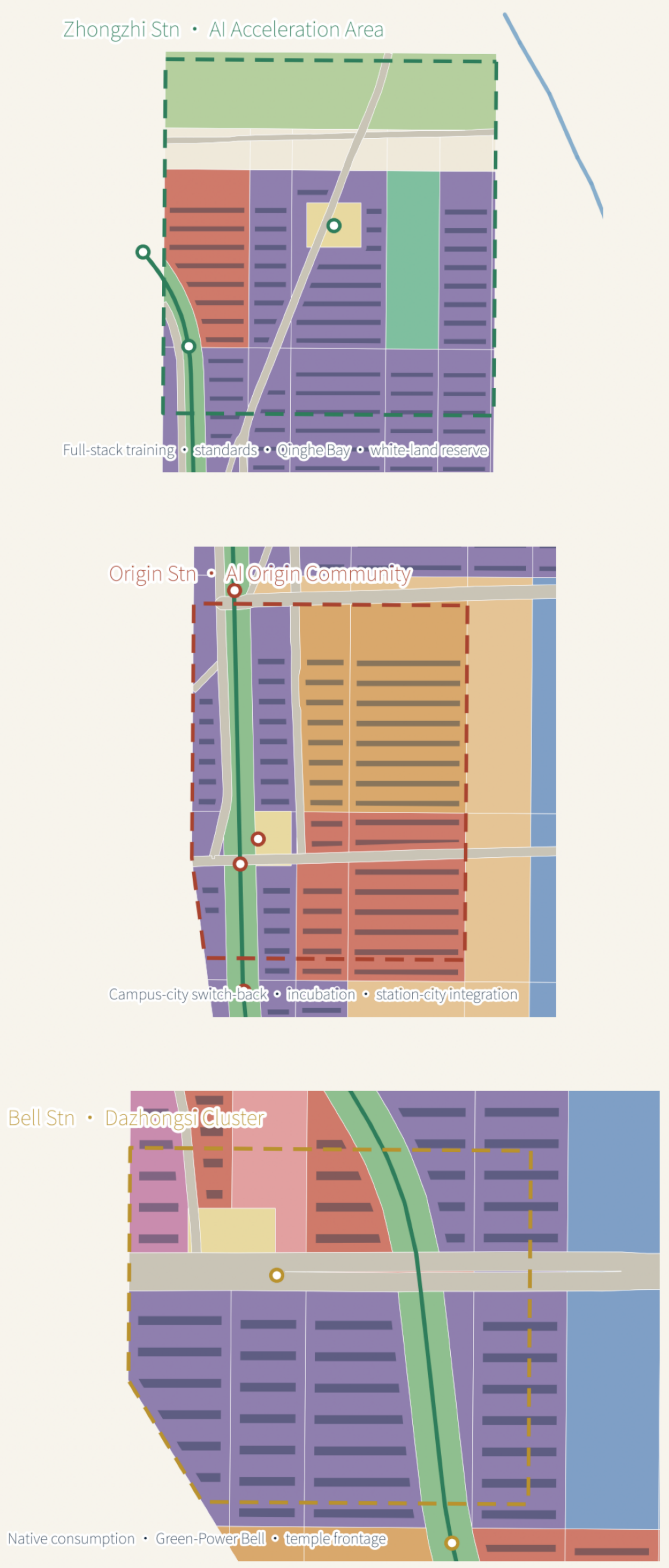
## Detailed Design of Three Key Areas

FIG-03

Zhongzhi (192.1 ha) • Origin (104.3 ha) • Bell (72.0 ha) — implementation-scheme depth (concept)

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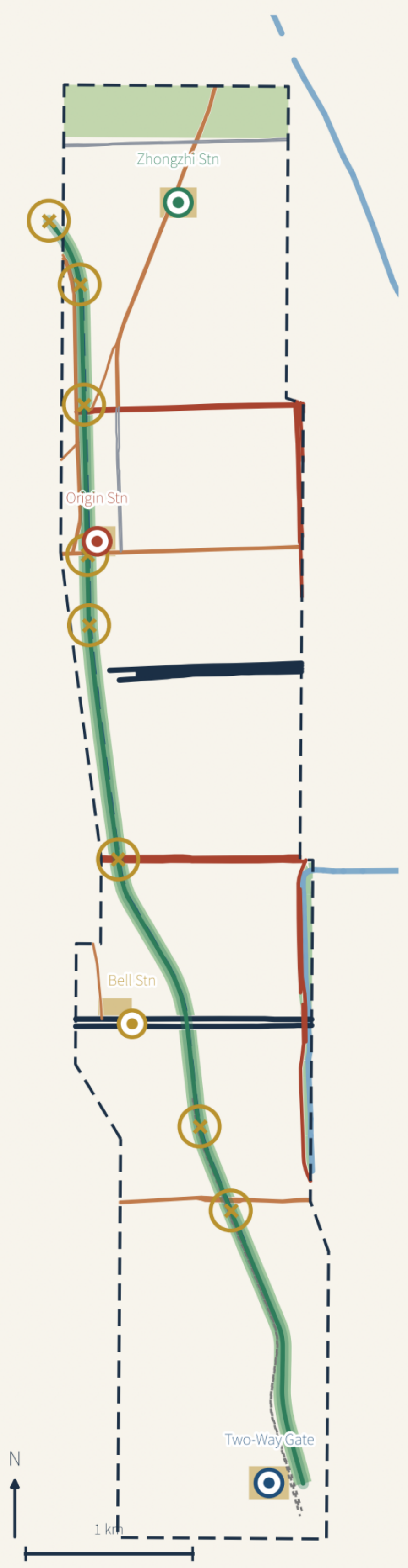
## Mobility & Blue-Green Systems

FIG-04

14 micro-circulation streets • through spine • 8 crossings • cooling loop • 4 station hubs

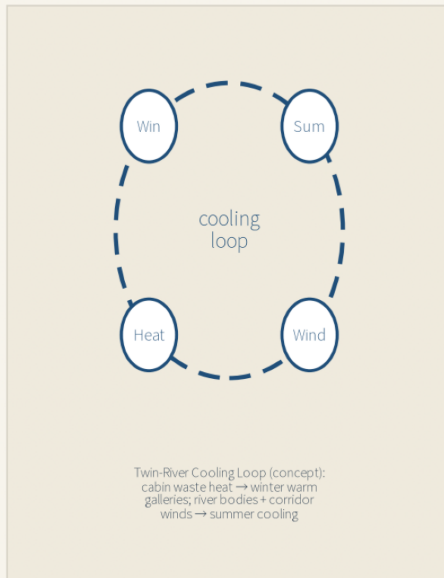



- Surface Line • 7.8 km heritage slow spine
- Under Line • Qinghuayuan HSR tunnel (indicative)
- Expressway
- Arterial
- Secondary
- Branch
- Crossings (8 east-west stitches)
- Qinghe / Xiaoyue R.
- Metro Line 13

Road hierarchy (measured in EPSG:4548)

| | | | |
|---|---|---|---|
| Expressway | 2 | 7.0 km | 44 m |
| Arterial | 5 | 12.6 km | 30-34 m |
| Secondary | 5 | 14.1 km | 22-24 m |
| Branch | 2 | 4.4 km | 16 m |
| Slow spine | 1 | 7.8 km | - |

Vehicular roads total 38.1 km on 74.1 ha (6.5% of the overall design area). This proposal adds no vehicular road and widens no existing redline; the entire capacity gain comes from the slow spine and the east-west stitching at the eight crossings.

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## Two-Way Protocol • a recomputable admission contract

Six rules taken line by line from this proposal, encoded as a machine-readable contract and a tabletop exercise; reviewers can run it themselves rather than take the text on trust.

| | |
|---|---|
| all three elements | no taking without returning |
| a named party can halt | privacy red lines |
| non-AI equivalent path | tests never called approved |

12 / 12 schedulable — Twelve service cards

6 / 6 caught by their rule — Deliberately broken runs

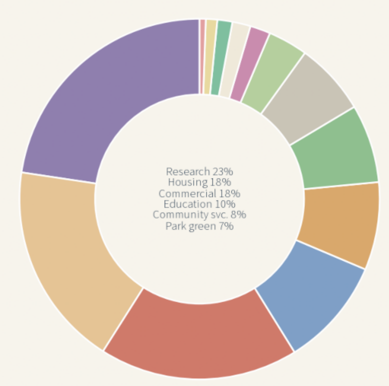
node visual/assets/run_twoway_tabletop.js • Node 18+, zero dependencies

## Metrics Recalculation & Evidence Chain

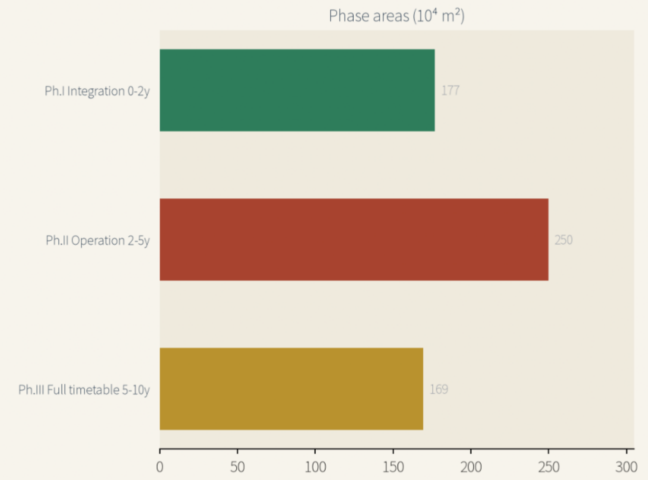
FIG-05

All spatial metrics recomputed from packaged GeoJSON in EPSG:4548 • spatial review PASS

Land use (full coverage 11,399,407 m²)

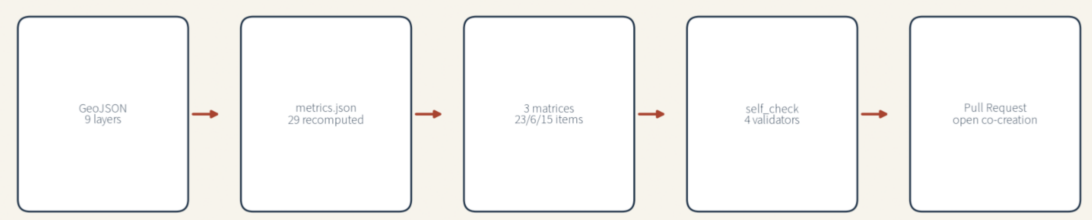



Phase areas ($10^4$ m²)




| | | |
|---|---|---|
| Site area<br>11,400,064 m²<br>+0.0006% vs announced | Key areas<br>3,683,997 m²<br>within ±0.01% | Green ratio<br>10.6%<br>corridor + two rivers |
| Public space<br>1.74%<br>timetable-multiplied | Concept GFA<br>5.89M m²<br>renewal potential ref. | Scenario cards<br>12 / 3 tests<br>Two-Way Protocol |

Evidence chain: every claim traces to layers and metrics




Pending official data (registered, never faked): FAR • max height • AI-index baseline • talent-density baseline

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Twelve run-cards (Two-Way Protocol: what goes up • what comes down • who can stop it)

| Run | Scenario | Node | Up-line | Down-line | Review & exit |
|---|---|---|---|---|---|
| T1 | Low-speed delivery test | Qinghe Bay | de-identified ops | free delivery hours | on-site stop |
| T2 | Traffic dispatch test | crossings | anonymous flows | published waits | manual fallback |
| T3 | Copilot sandbox | Bell Stn | authorized tickets | SME quota | human switchable |
| S4 | Cultural guide | mileposts | anonymous heat | 4-language guiding | human review |
| S5 | Health navigation | Xueyuan Rd | voluntary asks | elder escort | physician review |
| S6 | Safety ops review | the Gate | incident data | published drills | post-hoc only |
| S7 | Waste-heat dispatch | cooling loop | telemetry | warm gallery | manual bypass |
| S8 | Model release day | release hall | registration | open weights | community review |
| S9 | Accessible walking | 8 crossings | facility status | blind guidance | hotline backstop |
| S10 | Campus matching | Origin hall | licensed list | shared returns | manager review |
| S11 | Micro-climate | whole spine | sensors | heat warnings | stoppable points |
| S12 | Green-power chime | Bell plaza | power share | hourly chime | auditable source |